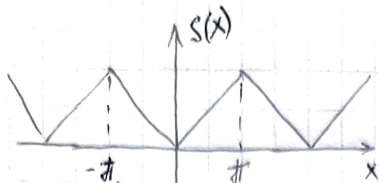


Консультация к экзамену от С.Д. Ивонтова

① $f(x) = |x| \quad [-\pi; \pi]$



сх. равн.

$$\{1, \sin nx, \cos nx\}_{n=1}^{\infty}$$

$b_n = 0$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx = \frac{2}{\pi n^2} \cos nx \Big|_0^{\pi} = \frac{2}{\pi n^2} ((-1)^n - 1)$$

$a_{2k} = 0$

$$a_{2k-1} = \frac{-4}{\pi (2k-1)^2}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \, dx = \frac{2}{\pi} \int_0^{\pi} x \, dx = \frac{2}{\pi} \cdot \frac{\pi^2}{2} = \pi$$

$$|x| \sim \frac{\pi}{2} + \sum_{k=1}^{\infty} \frac{-4}{\pi (2k-1)^2} \cos(2k-1)x$$

$$x=0 \Rightarrow 0 = \frac{\pi}{2} - \sum_{k=1}^{\infty} \frac{4}{\pi (2k-1)^2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$|x|$ - имеет с квадратом

Рав-во Парсеваля: $\frac{1}{\pi} \int_{-\pi}^{\pi} |x|^2 \, dx = \frac{a_0^2}{2} + \sum_{k=1}^{\infty} \frac{16}{\pi^2 (2k-1)^4} \rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4}$

то $\oplus$ график: $|x|' = \text{sign } x \approx \sum_{k=1}^{\infty} \frac{4}{\pi (2k-1)} \sin(2k-1)x$

$|x|$ - нечет. функ. $\rightarrow$ имеет

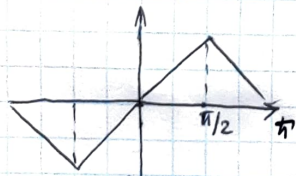
$$\int_0^x |t| \, dt = \frac{\pi}{2} x + \sum_{k=1}^{\infty} \frac{-4}{\pi (2k-1)^3} \frac{\sin(2k-1)x}{\sin(2k-1)x} \Big|_0^x \rightarrow x^2 - \pi x = \sum_{k=1}^{\infty} \frac{-8}{\pi (2k-1)^3} \sin(2k-1)x$$

Если бы было cos: $x^2 - \pi x = \sum_{k=1}^{\infty} \frac{-8}{\pi (2k-1)^3} \cos(2k-1)x - \sum_{k=1}^{\infty} \frac{-8}{\pi (2k-1)^3}$

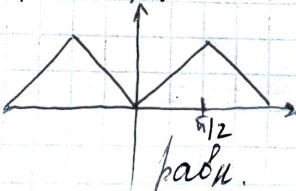
$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} (x^2 - \pi x) \, dx$$

② $f(x) = x \quad (0, \pi/2)$

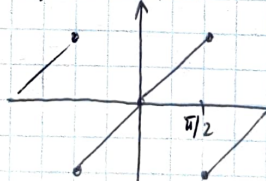
то $\sin(2n-1)x$:



то $\cos nx$:

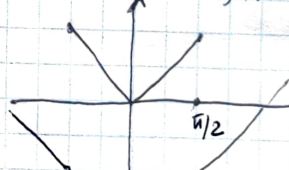


то $\sin 2nx$:



нет равн. эк.

то $\cos(2n-1)x$:



нет равн.

③ $f(x)$ - нечет. $[-\pi, \pi]$

Задача; 1,5 часа

$f(x) = -f(x-\pi)$ на $[0, \pi]$

$a_{2n} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos 2nx \, dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos 2nx \, dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos 2nx \, dx$

$x = t - \pi$

$\int_0^{\pi} \frac{f(t-\pi)}{-f(t)} \frac{\cos(2n(t-\pi))}{\cos 2nt} dt$

$a_{2n} = 0$

④ $f(x) = |x|(1-x^2)^2$ $[-1, 1]$

Найти кривую, найти кривую убывающую

$\{1, \sin nx, \cos nx\}$

$f(x) = \begin{cases} x(1-x^2)^2 & x \geq 0 \\ -x(1-x^2)^2 & x \leq 0 \end{cases}$

$f'(x) = \begin{cases} (1-x^2)^2 - 4x^2(1-x^2) & x \geq 0 \\ -(1-x^2)^2 + 4x^2(1-x^2) & x \leq 0 \end{cases}$

Разрыв в нуле

$k=1 \rightarrow a_n = o(\frac{1}{n})$

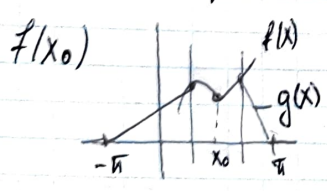
⑤ $\sum_{n=1}^{\infty} a_n \cos nx$ может ли быть ряд нечет. ф-ции, если $\sum_{n=1}^{\infty} |a_n|$ сход. ?

$|a_n \cos nx| \leq |a_n|$ - сход. $\Rightarrow$ ряд сход. равн. по В. $\Rightarrow$ для рядов с абсолютн. сходим.

⑥ $\sum_{n=1}^{\infty} a_n^2$ сход. $\rightarrow \sum_{n=1}^{\infty} \frac{|a_n|}{n}$ - сход.

$(|a_n| - \frac{1}{n})^2 \geq 0 \Rightarrow a_n^2 - \frac{2|a_n|}{n} + \frac{1}{n^2} \geq 0 \Rightarrow \frac{|a_n|}{n} \leq \frac{1}{2} (a_n^2 + \frac{1}{n^2})$

⑦ $f(x)$ - нечет. в окр. x_0 и ряд сход. в x_0 куда?

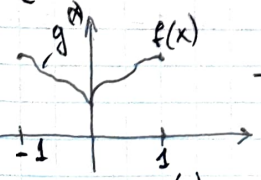


$\sigma_n(x)$ сходятся к $f(x)$ (где $g(x)$)

$\sum a_n$ - сход. к $A \Rightarrow \sigma_n \rightarrow A$ (если $f(x_0) = g(x_0) = A$)

⑧ $\{1, x^2, x^4, \dots\}$ в $C[0, 1]$ полна?

Полна?



$\forall f(x) \exists g(x) \in C[-1, 1]$

$g(x) = \sum_{n=0}^{\infty} a_n \cos nx$ $g(x)$ удовл. ⑦ Рунге

$\sigma_m(x) \Rightarrow g(x) \quad \forall \epsilon > 0 \exists m(x) : \|\sigma_m(x) - g(x)\| < \frac{\epsilon}{2}$

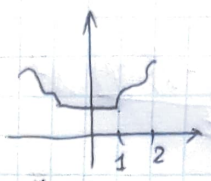
$\sigma_m(x) = \sum_{n=0}^m \lambda_n \cos nx \quad \cos nx = \sum_{k=0}^{\infty} (-1)^k (nx)^{2k} \cdot \frac{(-1)^k}{(2k)!}$ сход. равн. $[-1, 1]$

$$\|\sigma_m' - \sum_{k=0}^{m-1} M_k X^{2k}\| < \frac{\epsilon}{2}$$

$$\|g - \sigma_m'\| \leq \|g - \sigma_m\| + \|\sigma_m - \sigma_m'\| < \epsilon$$

9) $\{x^2, x^4, \dots\} \quad C[1, 2]$

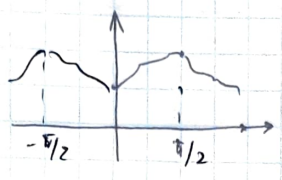
$g(x) = \frac{f(x)}{x^2} \quad C[1, 2]$



$\rightarrow \{1, x^2, x^4, \dots\} \dots \|g - \lambda_0 - \lambda_2 x^2 - \dots\| < \epsilon$

или: $t = \sqrt{x} \rightarrow$ Вейерштрасс

10) $\{1, \cos 2x, \cos 4x, \dots\} \quad C[0, \frac{\pi}{2}]$



2 задание

Равн. сходя.

11) $\int_0^\infty x^k \sin x^3 dx \quad (0, 1) = E_1$
 $(1, 2) = E_2$

$x^3 = t \quad x = t^{1/3}$
 $\int_0^\infty t^{1/3} \sin t \frac{dt}{3t^{2/3}} = \frac{1}{3} \int_0^\infty \frac{\sin t}{t^{1/3}} dt$

E_1 no Дирихле

1) $|\int_0^2 \sin t dt| \leq 2$

2) $\frac{1}{t^{2/3}} \leq \frac{1}{t^{2/3}} \rightarrow 0$ $\limsup_{t \rightarrow +\infty} \frac{1}{t^{2/3}} = 0$

3) $\frac{1}{t^{2/3}} -$ монот.

т.е. равн. no Дирихле

E_2 no Коши

$\int_{\delta_1}^{\delta_2} \frac{\sin t}{t^{2/3}} dt \approx \frac{1}{t^{2/3}} \int_{\delta_1}^{\delta_2} \sin t dt \approx \frac{2}{(2\pi n + \pi)^{2/3} - 2^{2/3}}$

$\exists \epsilon_0 = 1 : \forall B \in [1, +\infty) \exists \delta_1 = 2\pi[B] \neq \delta_2 = 2\pi[B] + \pi + 1 \exists \delta \in (\delta_1, \delta_2)$
 $|\int_{\delta_1}^{\delta_2}| \geq \epsilon_0$

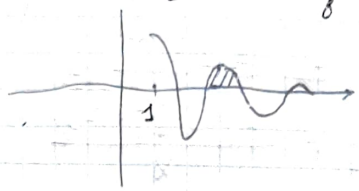
$\forall \epsilon > 0 \exists \delta_0 \in (1, 2) \forall \delta \in (\delta_0, 2) \rightarrow \left| \frac{2}{(2\pi n + \pi)^{2/3} - 2^{2/3}} - 2 \right| < \epsilon$
 $\frac{2}{(\quad)} \rightarrow 2 - \epsilon = 1 = 1 - \epsilon_0$ Не сходя. равн. (иногда сходятся, но Дирихле или Коши)

(12) $\int_1^{\infty} \frac{\cos x}{x} dx$ $E_1 = (0, 1)$
 $E_2 = (1, +\infty)$

E_2 - прав. по Дирихле $\int_1^2 |\cos x| dx \leq \frac{1}{2} \leq 1$
 $(0, 1)$: опр. е

$\forall \varepsilon > 0 \exists B \in (\frac{3\pi}{2}, +\infty) \forall \delta \in (B, +\infty) \forall \alpha \in E_1 \rightarrow \left| \int_{\alpha}^{\infty} \right| < \varepsilon$
 $\exists \varepsilon_0 = \dots \forall B \in (1, +\infty) \exists \delta = B \exists \alpha = \frac{3\pi}{2} \rightarrow \left| \int_{\alpha}^{\infty} \right| \geq \varepsilon_0$

$t_{\infty} = \alpha x$ (α - фикс.)
 $\left| \int_{\frac{2\pi}{3\pi/2}}^{\infty} \frac{\cos t}{t} dt \right| = \left| \int_{\frac{3\pi/2}{\alpha}}^{\infty} \frac{\cos t}{t} dt \right|$



(13) $\int_1^{\infty} \frac{\sqrt{1+\ln^2 x}}{x^3} dx$ $E_1 = (0, 1)$
 $E_2 = (1, +\infty)$

E_1 - прав. Б.

$|1 + \ln^2 x| \leq 1 + \ln x$ $\int_1^e + \int_e^{\infty}$
 $\ln x \geq 1 \quad x \geq e$

$\int_e^{\infty} \frac{\sqrt{1+\ln x}}{x^3} dx$

$\frac{\sqrt{1+\ln x}}{x^3} \sim_{x \rightarrow \infty} \frac{1}{x^3 \ln^{-1/2} x}$ эког.

E_2 - неравн. по Коши

(14) $\int_0^{\infty} \frac{1-e^{-\beta x}}{x} \cos x dx$ $\beta > 0$
 $\alpha > 0$ Вычислить кс. инт.

фикс α $\mathcal{I}(\beta)$ Форму. $\mathcal{I}'(\beta) = \int_0^{\infty} e^{-\beta x} \cos x dx =$
 $= \operatorname{Re} \int_0^{\infty} e^{-\beta x + i x} dx = \operatorname{Re} \frac{e^{-\beta x + i x}}{i - \beta} \Big|_0^{\infty} = \operatorname{Re} \frac{1}{\beta - i} = \frac{\beta}{\beta^2 + 1}$

$\mathcal{I}(\beta)^0 = \frac{1}{2} \ln(\beta^2 + 1) + C(\alpha)$

$\mathcal{I}(0) = \ln \alpha + C(\alpha) = 0$

1) f и f_{β} несп.

$F(t) = \begin{cases} \frac{1-e^{-t}}{t}, & t \neq 0 \\ 1, & t = 0 \end{cases}$ несп. на $\mathbb{R}$

$f(x, \beta) = \beta F(\beta x) \cos x$ несп. на $\mathbb{R}$

2) $\exists \beta : \mathcal{I}$ эког.

$\mathcal{I}(\beta) \in C(0, +\infty)$

$\mathcal{I}$ эког. б. по Дирихле: $\int_0^2 \cos x dx \leq \frac{1}{2}$

$\frac{1-e^{-\beta x}}{x}$ - монот. $\lim_{\beta \rightarrow \infty} \sup \left| \frac{1-e^{-\beta x}}{x} \right| \leq \lim_{x \rightarrow \infty} \frac{2}{x} = 0 \rightarrow \mathcal{I}(\beta)$ - несп.

3) Уст. or нроруб.

$$\left(\left[\beta_0, \beta_1 \right] \right)$$

$$\forall \beta \in (0, +\infty) |e^{-\beta x} \cos \alpha x| \leq e^{-\beta_0 x} \rightarrow \text{по } \beta. \text{ с.к. на } [\beta_0, \beta_1]$$

4) Константа: $\Gamma(0) = \lim_{\beta \rightarrow +0} \Gamma(\beta)$

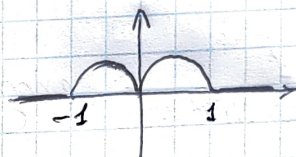
15) $f(x, \alpha) = \frac{e^{-\alpha x^2} - e^{-\beta x^2}}{x} \in \text{Уст. на } \mathbb{R}$
 $\lim_{\beta \rightarrow 0} f(x, \beta) = \frac{e^{-\alpha x^2} - 1}{x}$

$F(t) = \begin{cases} \frac{e^{-t} - 1}{t}, & t \neq 0 \\ 0, & t = 0 \end{cases}$ Уст. на $\mathbb{R}$

$f = \sqrt{\alpha} F(\sqrt{\alpha} x) - \sqrt{\beta} F(\sqrt{\beta} x)$
 $\alpha > 0, \beta > 0$

16) $f(x) = \begin{cases} \sin^2 \pi x, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$

Уст. Рубе



$a(y) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos xy \, dx$

$b(y) = 0$

$f(x) \sim \int_0^{\infty} a(y) \cos xy \, dy$

17) Уст. Рубе

$F[f](y) = F(y, f) = \frac{1}{\sqrt{2\pi}} \text{v.p.} \int_{-\infty}^{\infty} f(x) e^{-ixy} \, dx$
 $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} e^{-ixy} \, dx = 2 \int_0^{\infty} \frac{\sin^2 x}{x^2} \cos xy \, dx = \int_0^{\infty} \frac{\cos xy - \cos 2x \cos xy}{x^2} \, dx = \dots$
 Уст. Рубе

18) Уст. Рубе

$F(y, f^{(k)}) = (iy)^k F(y, f)$

$i^k F^{(k)}(y, f) = F(y, x^k f)$

$f, f', \dots, f^{(n)} \in C_L(\mathbb{R})$

$f(x) = x^2 e^{-\alpha x}$

$F(y, x^2 e^{-\alpha x}) = i^2 F''(y, e^{-\alpha x})$

$F(y, e^{-\alpha x}) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\alpha x} \cos xy \, dx = \sqrt{\frac{2}{\pi}} \frac{\alpha}{\alpha^2 + y^2}$

19) $f(x) = e^{-x^2} (2x^2 - 1) = -(x e^{-x^2})' = (e^{-x^2} - 2x^2 e^{-x^2})$

$F(y, f) = -iy F(y, x e^{-x^2}) = -iy \frac{i}{y} F'(y, e^{-x^2})$

$$(20) f = x^2 e^{-2|x|}$$

$F(y, f)$ бек. гурф.

$$F(y, f) = o\left(\frac{1}{y}\right) \quad y \rightarrow +\infty$$

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$$1) x^n x^2 e^{-2|x|} \in CL_1(\mathbb{R})$$

$$e^{-2|x|} \leq \frac{1}{|x|^m} \quad \forall m$$

$$x^{n+2} e^{-2|x|} \leq \frac{1}{x^2}$$

$$2) \begin{matrix} f''' \in C \\ f'' \in C \end{matrix}$$

$$F(y, f'') = (iy)^2 F(y, f)$$

$$\lim_{y \rightarrow \infty} F(y, f'') = 0 \Rightarrow o\left(\frac{1}{y}\right)$$

Обобщ. ф-лы

$$(21) \lim_{n \rightarrow \infty} n \cos nx \quad \delta \mathcal{D}'$$

$$\begin{aligned} \forall \varphi \in \mathcal{D} \quad \lim_{n \rightarrow \infty} (n \cos nx, \varphi) &= \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} n \cos nx \varphi(x) dx = \\ &= \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \varphi(x) d \sin nx = \lim_{n \rightarrow \infty} \left(\underbrace{\varphi(x) \sin nx}_{0 \text{ (ф.м.)}} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \varphi'(x) \sin nx dx \right) = 0 \end{aligned}$$

⑦ Рундана

$$|\varphi'| \leq M \quad \text{на } [-A, A]$$

$$\begin{aligned} (22) (\sin x \delta'''(x), \varphi) &= (\delta'''(x), \sin x \varphi) = -(\delta'(x), (\sin x \varphi)''') = \\ &= -(\delta, (\cos x \varphi + \sin x \varphi')'') = -(\delta, (-\cos x \varphi - \sin x \varphi' - 2 \sin x \varphi' + 2 \cos x \varphi'' + \cos x \varphi''' + \sin x \varphi''')) = \\ &= -(\underbrace{\varphi(0)}_{(\delta, \varphi)} + 3 \underbrace{\varphi''(0)}_{(\delta'', \varphi)}) = (\delta, \varphi) - 3(\delta'', \varphi) = \\ &= (\delta - 3\delta'', \varphi) \end{aligned}$$

$$\Rightarrow \sin x \delta''' = \delta - 3\delta''$$

Знарь!

$$F(y, \frac{1}{1+x^2}) = \sqrt{\frac{\pi}{2}} e^{-|y|}$$

мис. Ланнаса